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Research Article

Designing Catalyst Descriptors for Machine Learning in Oxidative Coupling of Methane

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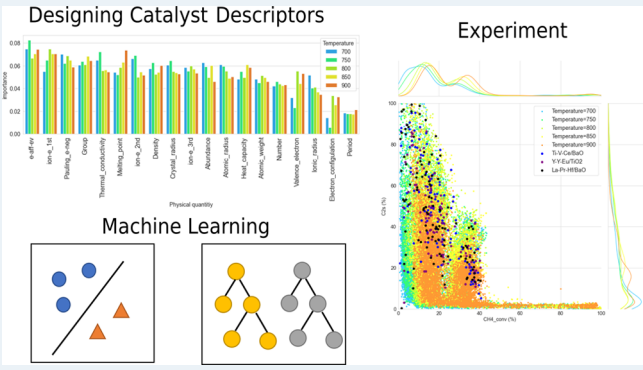
ABSTRACT: Catalysts descriptors for representing catalytic activities have been challenging in regard to machine learning. Machine learning and catalyst big data generated from high-throughput experiments are combined to explore the catalyst descriptors. Catalyst descriptors are designed using the physical quantities from the periodic table in the oxidative coupling of methane (OCM) reaction. Machine learning unveils the five key physical quantities representing ethylene/ethane selectivity (C_2s) in the OCM reaction, where machine learning predicted three catalysts to have high C_2s values. Experiments confirm that the proposed three catalysts have high C_2s values in the OCM reaction. Hence, the physical quantities can be used as alternative descriptors for designing heterogeneous catalysts.



KEYWORDS: *machine learning, catalyst informatics, descriptors, oxidative coupling of methane, methane oxidation*

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INTRODUCTION

Machine learning has had a large impact in catalyst design, where catalysts are designed from the trends and patterns found in catalyst data.[1−4] The idea of supervised machine learning is to essentially solve $y = f(x)$, where y and x are described as objective and descriptor variables, respectively.[5−7] However, the challenge in machine learning lies with descriptor variables, where catalyst descriptors for predicting catalytic activities still remain a mystery.[8−10] If catalyst descriptors can be determined, one can consider that further accurate machine learning models can be designed in principle. Here, catalyst descriptors are explored with the use of physical quantities from the periodic table.

The oxidative coupling of methane (OCM) reaction is selected as a target reaction. The OCM reaction involves the direct conversion of methane (CH_4) to ethylene/ethane (C_2H_4/C_2H_6).[11−13] Machine learning has been implemented to search for catalysts in the OCM reaction where the C_2 yield is set as an objective variable.[14] There are two types of descriptor variables in the OCM reaction, where one type is of the process conditions and the other is of catalyst information. Process conditions such as temperature and gas flow are treated as numerical variables; therefore, those variables can be directly trained by machine learning. On the other hand, catalyst compositions are generally expressed as atomic symbols and therefore must be converted to numerical variables for machine learning. In the OCM reaction, catalysts have been converted into atomic numbers as well as binary variables using one hot encoding.[14−16] However, descriptors for determining C_2 yield remains uncertain; thus, physical quantities from the periodic table should be explored. Here, catalyst descriptors for C_2H_4/C_2H_6 selectivity in the OCM reaction are performed using machine learning and the information from the periodic table, after which the designed catalysts are then experimentally tested.

DATA PREPROCESSING AND MACHINE LEARNING

Oxidative coupling of methane catalyst big data is used where data that have been previously generated through high-throughput experiments are utilized.[17] The data set consists of the following descriptor variables: atomic elements M1, M2, and M3 from catalyst compositions consisting of M1-M2-M3, support, temperature (°C), CH_4 flow (mL/min), O_2 flow (mL/min), and Ar flow (mL/min). CH_4 conversion, C_2 yield, and the selectivities and yields of C_2, C_2H_4, C_2H_4, CO, and CO_2 are also included. It has been reported that the data set contains some outliers due to carbon deposition on catalysts.[17]

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Therefore, outlier data are removed based on the following rules: data that fall out of ranges $0 \leq CH_4$ conversion ≤ 100 and selectivity(C_2H_6, C_2H_4, COs, and CO_2) ≤ 100). A total of 31334 data are used after the outlier data are removed. Furthermore, 19 physical quantities of the periodic table are assigned to M1, M2, and M3, where the average is taken. In particular, the following 19 descriptors are used: atomic number (Number), atomic weight (Atomic_weight), electron affinity (e-aff-ev), Pauling electronegativity (Pauling_e-neg), group in periodic table (Group), period in periodic table (Period), the number of valence electron (Valence_electron), atomic radius (atomic_radius), ionic radius (Ionic_radius), crystal radius (Crystal_radius), electron configuration (Electron_configuration), first ionization energy (ion-e_1st), second ionization energy (ion-e_2nd), third ionization energy (ion-e_3rd), abundance of the chemical elements (Abundance), thermal conductivity (Thermal_conductivity), melting point (Melting_point), density (Density), and heat capacity (Heat_capacity).[18] Note that all data are collected in the Supporting Information.

Both supervised and unsupervised machine learning is used to explore the data. Random forest classifier (RFC) and support vector classifier (SVC) are implemented within scikit-learn.[19] The number of trees for RFC is set to 100. Hierarchical clustering is implemented using scipy. Cross-validation is performed where data are randomly split into 20% test and 80% trained data. The average score of 10 random test and trained data in cross-validation is evaluated.

■ EXPERIMENTAL METHODS

Three catalysts predicted by machine learning—Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO—were tested in experiments. A support powder was impregnated with a solution of the specified metal precursor at 50 °C for 6 h. The resultant solid was vacuum-dried and subjected to calcination at 1000 °C under air for 3 h. The reagents are described in a previous report.[17] In the case of Pr, which is not included in the original data, $Pr(NO_3)_3 \cdot 6H_2O$ was used as a precursor. The amount of each element was fixed at 0.37 mmol/g of support, as in the original data.[17] A homemade high-throughput catalyst evaluation system was used, which can evaluate the performance of 20 catalysts in a fixed-bed arrangement under a programmed series of reaction conditions.[20,21] A fixed volume of crushed catalyst powder was loaded into a quartz reaction tube (10 mm height in a 4 mm i.d. quartz tube), and the performance of each catalyst was acquired under 135 reaction conditions, differing in the reaction temperature (700, 750, 800, 850, 900 °C), the total flow volume (10, 15, 20 mL/(min channel)), the CH_4/O_2 ratio (2, 4, 6 mol/mol), and the Ar concentration (P_{Ar} = 0.15, 0.40, 0.70 atm). The performance of a catalyst is represented by the condition that led to the maximum C_2 yield out of the 135 conditions. Further details are as described in previous reports.[17,22]

■ RESULTS AND DISCUSSION

Data visualization is performed in order to understand the trends and patterns that lie within catalyst big data. In particular, C_2 selectivity (C_2s) and CH_4 conversion are visualized using a scatter plot such as that shown in Figure 1, where the color is assigned based on the following temperatures: 700, 750, 800, 850, and 900 °C. Figure 1 illustrates that each temperature displays unique trends where

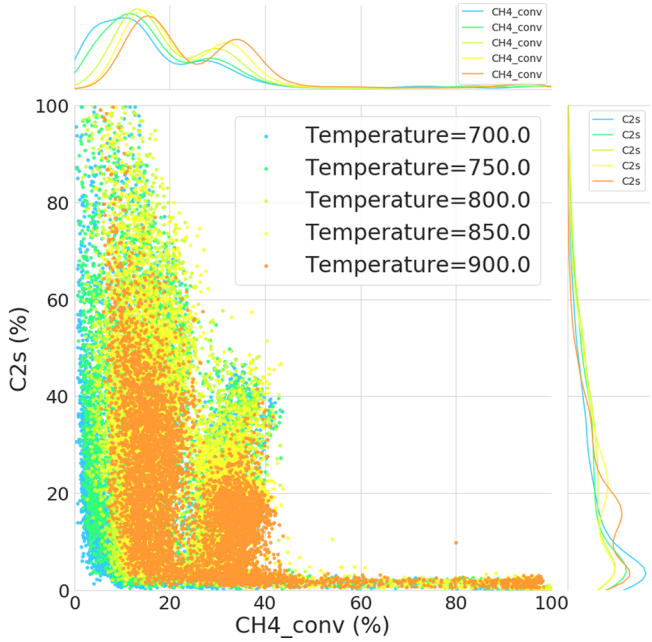

Figure 1. C_2 selectivity and CH_4 conversion. The color represents the temperature. Data frequencies are also drawn.

data can be classified into three groups for each temperature, as there are low, medium, and high C_2s groups in Figure 1. For instance, some catalysts have very low C_2 while some have high C_2. Hierarchical clustering is performed in order to unveil hidden groups for each temperature, as shown in Figure 2a where C_2s is used as an objective variable. Data at each temperature are classified into three groups based on the distance in the dendrogram. Figure 2b demonstrates that C_2s is classified into three groups: low, medium, and high C_2s. Here, one can consider that these three classified groups can be treated as objective variables for supervised classification modeling.

A random forest classifier (RFC) is implemented where three RFC models are created for each temperature. Nineteen descriptor variables for catalyst elements and the three groups revealed by the hierarchical clustering are set as descriptors and objective variables, respectively. An importance analysis by RFC is performed. Figure 3a shows that the following eight descriptor variables have high importance: electron affinity (e-aff-ev), the first ionization energy (ion-e first), Pauling electronegativity (Pauling e-neg), Group, thermal conductivity (Thermal conductivity), Melting point (Melting point), the second ionization energy (ion-e second), and the density. Furthermore, a second round of importance analysis is performed among the top eight descriptors presented in Figure 3a, where the objective variable is also set to C_2s. Here, the following top five descriptors shown in Figure 3b are selected as highly important descriptors for predicting C_2s: ion-e first, ion-e second, e-add-ev, Pauling e-neg, and density. Given that the support also plays an important role, one hot encoded support information is also included as descriptors with the chosen five descriptors. The importance of the five descriptors with support information is presented in Figure 3c, which illustrates that five physical quantities still have a major contribution toward determining C_2s. Cross-validation is also performed with the selected five descriptors with and without support, where both RFC and SVC are used. The scores for

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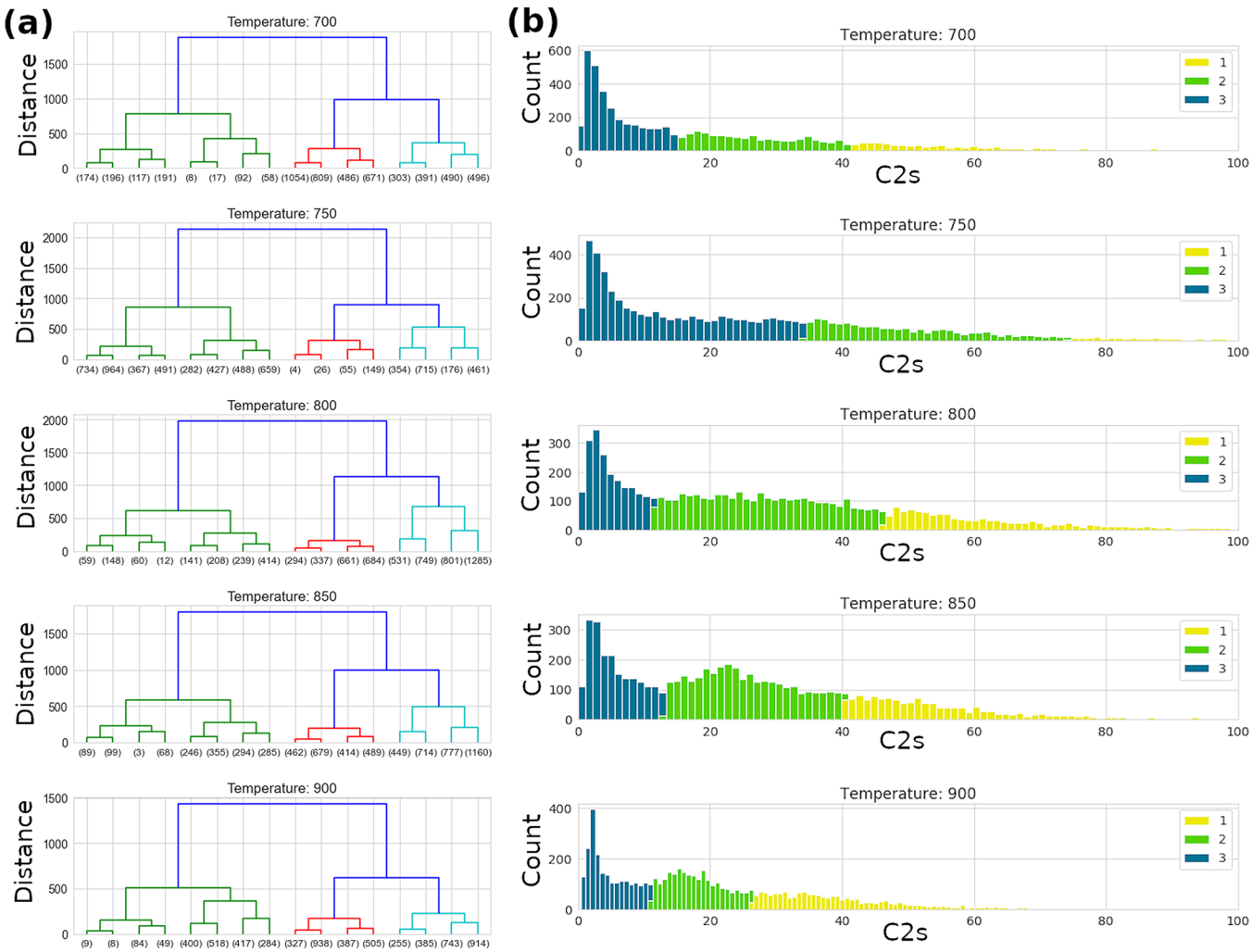

Figure 2. (a) Dendrogram created by hierarchical clustering. (b) Classified data by hierarchical clustering at each temperature.

each model are collected in Table 1. Here, one can see that the scores are within the range of 0.67−0.84, thereby showing that accurate machine-learning models are created. More importantly, catalyst descriptors for C_2 selectivity are successfully determined without using experimental conditions.

An inverse analysis is performed to predict the catalysts having high C_2s using trained RFC and SVC models. In particular, the 14 descriptors shown in Figure 3c are used as descriptor variables, while the C_2s group shown in Figure 2b is set as an objective variable. The following six machine-learning models are used for prediction: RFC and SVC with data at 700, 800, and 900 °C. Here, catalysts are designed following the structure "M1-M2-M3", where M1, M2, and M3 represent different sets of atomic elements and every possible combination of M1, M2, and M3 elements is considered. Here, M1, M2, and M3 combinations are created using the following 71 elements: Li, Be, B, C, N, O, F, Na, Al, Si, P, S, Cl, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Ga, Ge, As, Se, Br, Rb, Sr, Y, Zr, Nb, Mo, Ru, Rh, Pd, Ag, In, Sn, Sb, Te, I, Xe, Cs, Ba, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Lu, Hf, Ta, W, Re, Os, Ir, Pt, Au, Tl, Pb, and Bi. Once the combinations are made, the averages of ion-e first, ion-e second, e-add-ev, Pauling e-neg, and density are taken and used as descriptors. Thus, a total of 62196 element combinations are considered based on permutations of three elements available from the 71 elements.

To start, prediction of C_2s for 6 models is performed using five descriptors shown in Figure 3b which do not include support information. In particular, 3 RF and 3 SVR models for 700, 800, and 900 °C are used. The 62196 element combinations are then given to each of the 6 machine learning models where the corresponding C_2s groups are returned. Among the results from different machine-learning models, catalysts that appeared in all machine-learning models with high C_2s group are collected in Table 2, where 15 catalysts are proposed. However, given that the support plays an important role in the OCM reaction, appropriate supports for the listed 15 catalysts are then sought for use in machine learning. In particular, the 15 catalysts are then given to the 6 machine learning models, which are trained using the 14 descriptors shown in Figure 3c. Nine supports, as shown in Figure 3c, with the 15 selected catalysts are given to the 6 trained machine learning models. As a result, a total of 21 catalysts with supports are predicted to have high C_2s in all of the 6 trained machine-learning models, as shown in Table 2. More specifically, the following 4 previously unreported catalysts are predicted to have high C_2s: Ti-V-Ce-BaO, Y-Y-Eu-SiO_2, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO. The predicted catalysts are then validated via experiments.

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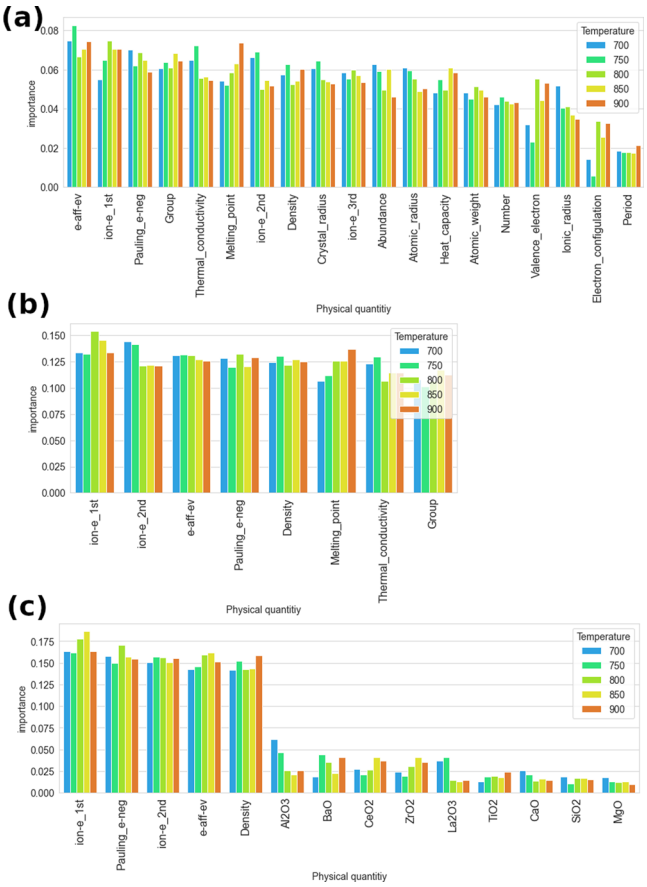

Figure 3. Importance analysis by random forest classification: (a) 18 physical properties without support; (b) second round of importance analysis of top 8 descriptors from (a); (c) top 5 descriptors with support.

Table 1. Cross-Validation Score Using RFC and SVC[a]

temp (°C)	RFC	SVC
700	0.786	0.784 (C = 5, gamma = 0.1)
750	0.841	0.841 (C = 4, gamma = 0.1)
800	0.766	0.765 (C = 4, gamma = 0.1)
850	0.718	0.718 (C = 4, gamma = 0.1)
900	0.672	0.674 (C = 4, gamma = 0.1)
temp (°C)	RFC (support)	SVC (support)
700	0.792	0.790 (C = 5, gamma = 0.1)
750	0.844	0.844 (C = 4, gamma = 0.1)
800	0.773	0.772 (C = 4, gamma = 0.1)
850	0.724	0.724 (C = 4, gamma = 0.1)
900	0.679	0.680 (C = 4, gamma = 0.1)

[a]The top and bottom parts of the table represent five descriptors and five descriptors with a support, respectively. For SVR, hyper-parameters C and gamma values are given.

The predicted catalysts are validated using high-throughput experiments. In particular, Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO are experimentally tested. Note that Y-Y-Eu-SiO_2 is not performed in the experiments, as the element combination is similar to that of Y-Y-Eu-TiO_2. The best C_2y values with the corresponding experimental conditions of these three catalysts are given in Table 3. From Table 3, one can see that La-Pr-Hf-BaO has a particularly high C_2s of 84% with a C_2 yield of 15.6%. Further analysis is performed in order to better understand the C_2s of the three catalysts. Here, CH_4 conversion and C_2s of Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO against various experimental conditions are shown in Figure 4. Note that all experimental data are collected in the Supporting Information. Figure 4 shows that Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO tend to have high C_2s. More importantly, these catalysts are confirmed to have high C_2s, as shown in Figure 4. Note that these catalyst reactions retain their activity and selectivity during the catalyst test. It should also be pointed out that La-Pr-Hf-BaO offers a major advancement in OCM catalysts, as La-Pr-Hf-BaO has a C_2 selectivity of 84% with a C_2 yield of 15.6%. Previously, there have only been four catalysts that were reported to have high C_2y values with high C_2s, where all four of these catalysts use a MgO support.[23] In this work, we have found that a new support, BaO, coupled with La, Pr, and Hf can reach such high activity, thus creating a new class of catalysts. Hence, high-C_2s catalysts are found to have been successfully designed using machine learning and the physical quantities as catalyst descriptors.

Table 2. Predicted Catalysts by Six Machine Learning Models[a]

A	B	C	D
Li-Zr-Cs	Ti-V-Ce-BaO	Na-Eu-W-BaO	Li-Zr-Cs-BaO
Na-Sr-Mo	Y-Y-Eu-SiO_2	Ti-Sr-La-MgO	Na-Sr-Mo-SiO_2
Na-Mo-Sm	Y-Y-Eu-TiO_2	Ti-Sr-La-TiO_2	Na-Eu-W-ZrO_2
Na-Eu-W	La-Pr-Hf-BaO	Mo-Ba-Ce-SiO_2	K-V-Mo-BaO
K-V-Mo		Mo-Ba-Ce-BaO	K-Zr-La-BaO
K-Zr-La		Ce-Nd-Hf-SiO_2	Ti-Sr-La-SiO_2
Ti-V-Ce			V-Zr-Eu-BaO
Ti-Sr-La			Mo-Cs-W-BaO
V-Zr-Eu			Mo-Ba-Ce-TiO_2
Y-Y-Eu			Ce-Nd-Hf-BaO
Mo-Cs-W			Tb-Hf-W-La_2O_3
Mo-Ba-Ce			
La-Pr-Hf			
Ce-Nd-Hf			
Tb-Hf-W			

[a]Catalysts that appeared in all machine-learning models as high C_2s group are collected: (A) prediction without support; (B) unreported catalysts with support; (C) catalysts that were previously reported but with an unreported catalyst and support combination; (D) reported catalysts in training data.

Table 3. Experimental Results of Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO[a]

catalyst	CH_4 con	C_2y	C_2s	T (°C)	CH_4/O_2
La-Pr-Hf-BaO	18.50	15.57	84.14	800	2
Ti-V-Ce-BaO	26.02	17.63	67.75	850	4
Y-Y-Eu-TiO_2	37.63	15.62	41.52	900	2

[a]CH_4 con, C_2y, and C_2s represent CH_4 conversion (%), C2 yield (%), and C2 selectivity (%), respectively. T and CH_4/O_2 represent temperature and CH_4/O_2 ratio, respectively. Note that the best C_2y values with corresponding experimental conditions are given.

The physical meaning of the following discovered five descriptors are investigated: ion-e first, ion-e second, e-add-ev, Pauling e-neg, and density. In particular, the average densities, electronegativities, electron affinities, and first and second ionization energies of predicted high-C_2-selectivity catalysts shown in Table 2B−D are plotted as shown in Figure 5 where the average of 5 descriptors in M1, M2, and M3 created from 71 elements are taken for comparison. Figure 5 shows that

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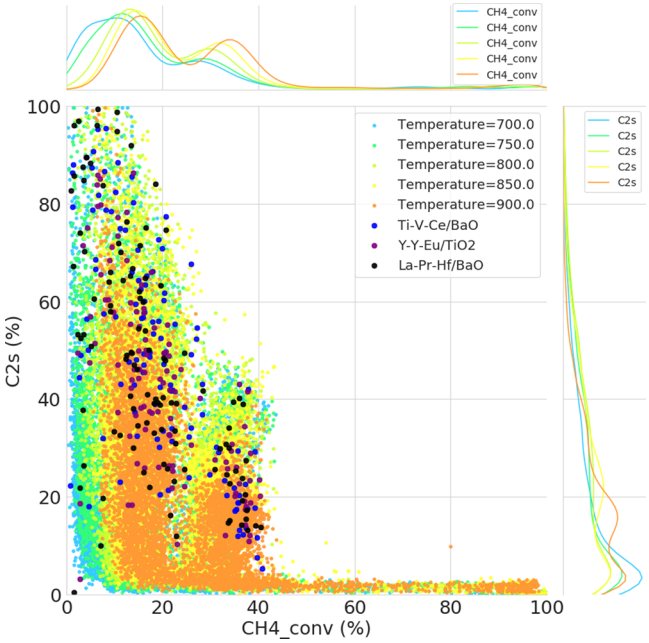

Figure 4. CH_4 conversion and C_2s of Ti-V-Ce-BaO, Y-Y-Eu-TiO_2,and La-Pr-Hf-BaO against various experimental conditions. The color represents the temperature.

predicted high-C_2s catalysts tend to have low electronegativities, electron affinities and first ionization energies compared to the average variables. One can consider that a low electronegativity, electron affinity, and first ionization energy might be key to the resulting high C_2 selectivity. This finding was further investigated in previous works. They suggested that catalysts with low ionization enthalpies and low electronegativities might modify the structure of host oxides; thus, it was suggested to improve C_2 selectivity.[23] Thus, descriptors found in this work have good agreement with observations in experiments.

■ CONCLUSION

Catalyst big data and machine learning are combined to search the catalyst descriptors in the oxidative coupling of the methane reaction. In particular, physical quantities from the periodic table are implemented to design catalyst descriptors for predicting C_2 selectivity in the OCM reaction. The following physical quantities are discovered to have an impact on C_2 selectivity: the first ionization energy, the second ionization energy, electron affinity, Pauling electronegativity, and density. With RFC and SVC, Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO are predicted to have high C_2s. These catalysts are tested in experiments, and Ti-V-Ce-BaO, Y-Y-Eu-TiO_2, and La-Pr-Hf-BaO are found to result in high C_2s. Thus, physical quantities from the periodic table are found to link to catalytic activity when they are used as descriptors through

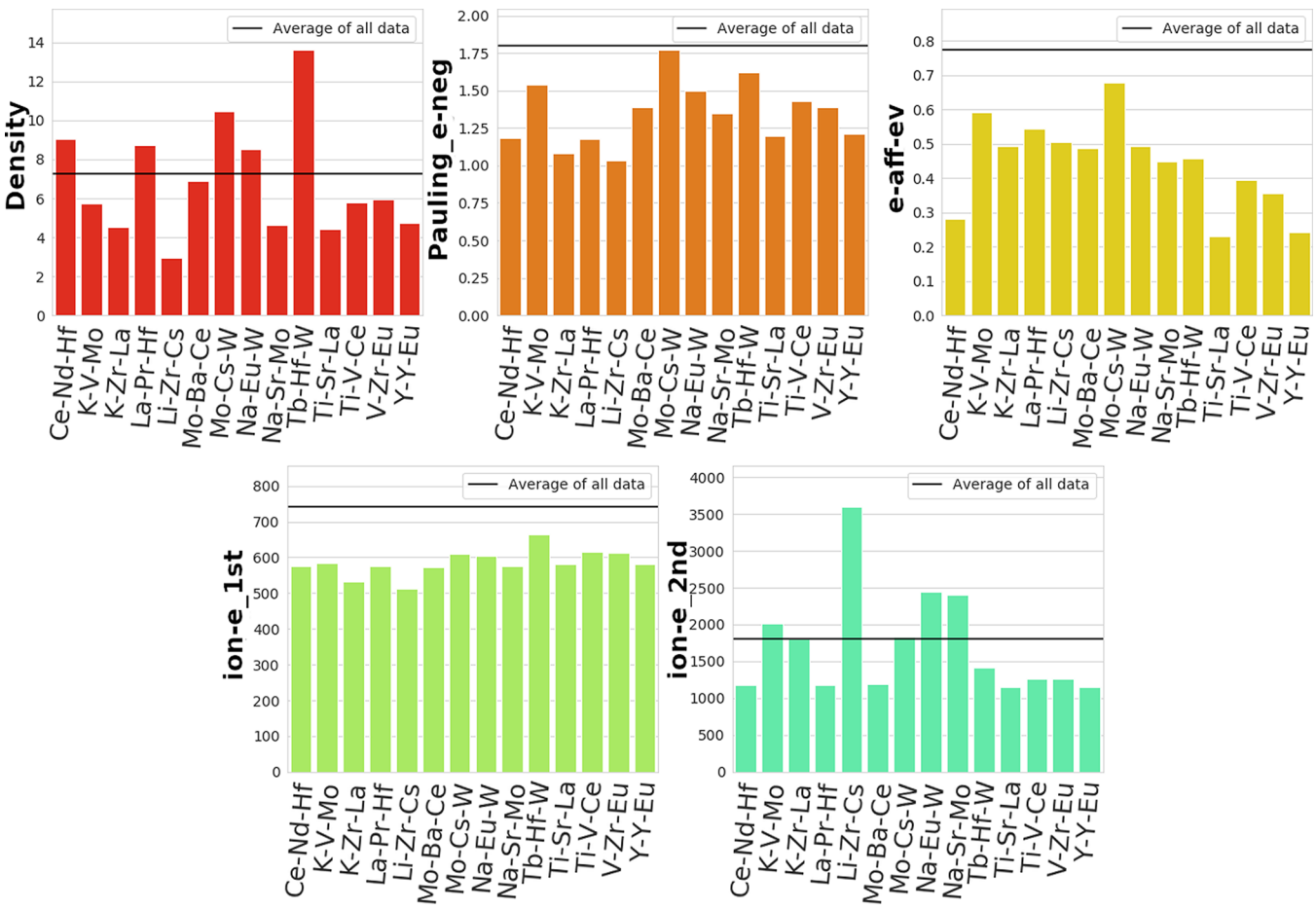

Figure 5. Average density, electronegativity, electron affinity, and first and second ionization energies of predicted high-C-selectivity catalysts. The average of 5 descriptors in M1, M2, and M3 created from 71 elements are taken for comparison.

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machine learning, leading to alternative descriptor choices for the design of heterogeneous catalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.2c03142.

Training data for machine learning and experimental validation data (ZIP)

Further details regarding the data (PDF)

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Notes

The authors declare no competing financial interest.

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